

```

import numpy as np
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input

x = np.array([[0,0], [0,1], [1,0], [1,1]])
y = np.array([[0], [1], [1], [0]])

model = Sequential([
    Input(shape=(2,)),          # New Input layer here
    Dense(4, activation='relu'),
    Dense(1, activation='sigmoid')
])

model.compile(loss='binary_crossentropy', optimizer='adam',
metrics=['accuracy'])

model.fit(x, y, epochs=1000, verbose=0)

prediction = model.predict(x)
print("Predictions:")
print(np.round(prediction))

1/1 ————— 0s 102ms/step
Predictions:
[[1.]
 [1.]
 [1.]
 [0.]]

```